

Coral-Inspired Zinc Acrylate Polymer Utilizing Coumarin as the Fluorescent Unit for Marine Antifouling

Jianwei Hou, Guobo Chen, Xinghai Hao, Jiali Xu, Geoffrey I. N. Waterhouse, Zhiming Zhang,* and Liangmin Yu*



Cite This: *Biomacromolecules* 2025, 26, 1799–1815



Read Online

ACCESS |



Metrics & More

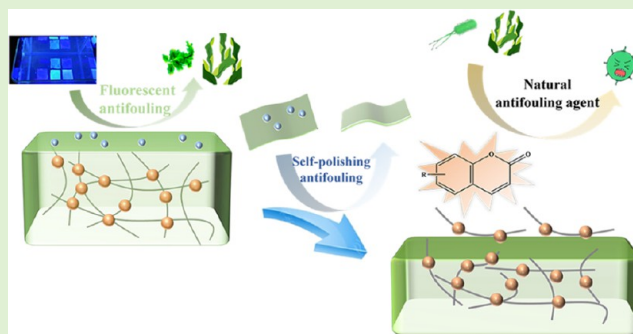


Article Recommendations



Supporting Information

ABSTRACT: The need for low-cost and effective antifouling solutions drives innovation in the fields of chemistry, materials science, and biology. In this work, guided by the antifouling strategies used by fluorescent corals, a series of fluorescent zinc acrylate polymer coatings containing coumarin units (ZAR-coumarin) was successfully prepared. The ZAR-coumarin coatings demonstrated excellent antifouling properties due to the synergistic action of multiple antifouling mechanisms, including fluorescent antifouling, natural bactericidal activity, and self-polishing surface renewal (due to ester group (–COO–Zn–OOC–) cleavage). Compared with zinc acrylate coatings without coumarin units (ZAR), the introduction of coumarin units significantly improved the inhibition efficiency for both bacteria and algae. In marine environment tests, the ZAR-AMCO-1, ZAR-ADMCO-1, and ZAR-CAMCO-1 coatings containing optimized amounts of different types of coumarin units maintained good antifouling properties over a 160-day field test period. This research presents an innovative approach to creating marine antifouling coatings.



1. INTRODUCTION

Marine biofouling, involving the accumulation of organisms, such as algae, barnacles, and mollusks, on submerged surfaces, impacts the performance and lifetime of both stationary maritime structures and vessels. Economic costs associated with marine biofouling (including its prevention or removal) are estimated to be \$15 billion per year for the global marine industry.^{1,2} Further, marine biofouling raises concerns regarding biosecurity and environmental degradation. Presently, applying antifouling coatings is considered to be the most cost-effective antifouling method. Traditional antifouling coatings used toxic biocides, such as tributyltin (TBT) and copper-based compounds, which have raised environmental concerns and regulatory scrutiny due to their detrimental effects on nontarget organisms and marine ecosystems.^{3,4} In recent years, there has been a shift toward sustainable antifouling coatings,⁵ including low surface energy antifouling coatings,^{6,7} tin-free self-polishing antifouling coatings,^{8–11} biodegradable coatings,¹² hydrogel coatings,^{12–14} and bioinspired antifouling coatings.^{15–17} However, antifouling coatings with a single function are somewhat limited in their usefulness in seawater. For example, low surface energy antifouling coatings and self-polishing antifouling coatings have poor antifouling ability in static environments. Therefore, it is necessary to develop new antifouling coatings with multiple antifouling mechanisms. Bioinspired antifouling coatings are currently attracting a great deal of attention, mimicking natural surfaces and chemical functions used by living

organisms. Notable examples include the micro/nanostructured surfaces used by sharks,¹⁸ mucus-like hydrogels similar to epidermal mucus,¹⁹ slippery liquid-infused porous surfaces (SLIPS) inspired by earthworms,²⁰ etc. Bioinspired antifouling coatings have good environmental compatibility, broad development prospects, and practical value. However, bioinspired antifouling coatings with a single function are not well-suited to meet the antifouling requirements in harsh marine environments. Designing bioinspired antifouling coatings incorporating other antifouling properties is therefore a promising strategy to further improve marine antifouling performance.

Recently, it has been found that certain corals emit weak fluorescence, which can promote the photosynthesis of *Cordyceps sinensis* and effectively inhibit the sedimentation of algae, thus achieving efficient antifouling.²¹ Inspired by this, fluorescent antifouling is gaining attention.^{21–24} For example, sandwiched poly(dimethylsiloxane) (PDMS)-phosphorus–silicone rubber coatings were fabricated and exhibited broad-spectrum antifouling performance.²⁵ As a class of naturally occurring compounds, coumarin derivatives with an electron-

Received: November 11, 2024

Revised: February 5, 2025

Accepted: February 6, 2025

Published: February 17, 2025

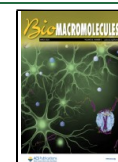


Table 1. Synthesis of Coumarin Analogs Containing Hydroxyl Groups

Synthesis scheme of coumarin containing hydroxyl structure

Phenolic compound	Catalyst	Coumarins with hydroxyl groups
Resorcinol 	ZrCl ₄	7-hydroxy-4-methylcoumarin (HMCO)
2-methylresorcinol 	ZrCl ₄	7-hydroxy-4,8-dimethylcoumarin (HDMCO)
Phloroglucinol 	ZrCl ₄	5,7-dihydroxy-4-methylcoumarin (5,7-DHMC0)
Pyrogallol 	H ₂ SO ₄	7,8-dihydroxy-4-methylcoumarin (7,8-DHMC0)
4-chlororesorcinol 	H ₂ SO ₄	6-chloro-7-hydroxy-4-methylcoumarin (CHMCO)

donating groups at the 7-position and an electron-withdrawing group at the 3-position (or 4-position) show strong fluorescence.²⁶ Furthermore, coumarin compounds can interact with a wide range of enzymes and receptors in organisms, thus showing many pharmacological properties, such as antioxidant, antibacterial, antifungal, antiviral, and anticancer activities.²⁷ Therefore, coumarin compounds are promising natural antifouling agents, with their use in antifouling coatings being fully consistent with the “from nature to nature” principle.^{28,29}

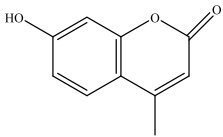
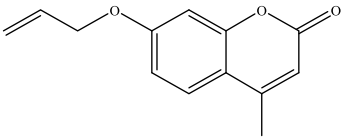
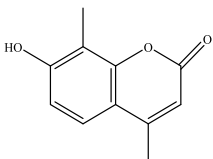
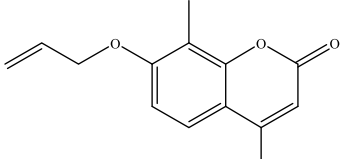
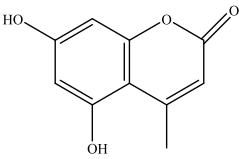
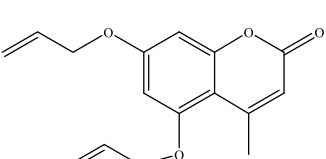
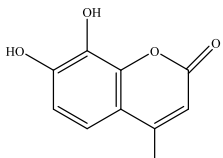
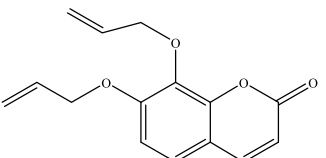
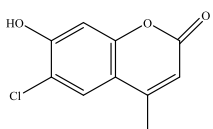
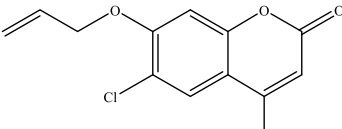
In this work, various fluorescent coumarin derivatives were grafted into a self-polishing zinc acrylate resin (ZAR) by the copolymerization of acrylic ester (ethyl acrylate and butyl acrylate), acrylic acid, and fluorescent coumarin derivatives. The antifouling performance of the ZAR-coumarin coatings was then evaluated, with the goal of synergistically exploiting multiple antifouling mechanisms (fluorescence, antimicrobial, and self-polishing). In agreement with our hypothesis, the ZAR-

coumarin coatings show a highly effective antifouling performance. Real marine fouling tests demonstrated that ZAR-coumarin provided 160 days of fouling resistance during the peak seasons of microbial growth.

2. EXPERIMENTAL PARTS

2.1. Chemicals and Materials. 2,2'-Azobis(2-methylpropionitrile) (AIBN), resorcinol, 2-methylresorcinol, phloroglucinol, pyrogallol, 4-chlororesorcinol, zirconium chloride (ZrCl₄), H₂SO₄, ethyl acetoacetate, ethanol, acetone, KI, 3-bromopropene, NaOH, nutrient agar, and K₂CO₃ were obtained from Sinopharm Chemical Co., Ltd. All of the chemicals were of analytical reagent (AR) grade and were used as received without purification. Xylene, *n*-butanol, ethyl acrylate (EA), butyl acrylate (BA), acrylic acid (AA), butyl acetate, ZnO, and benzoic acid were all industrial-grade chemicals and used without purification. A laboratory RO-DI water purification apparatus was used to obtain distilled water.

Table 2. Synthesis of Coumarin Compounds Containing Allyl Groups

Coumarin derivative with hydroxyl groups	Coumarin derivative with allyl groups
7-hydroxy-4-methylcoumarin (HMCO)	7-allyloxy-4-methylcoumarin (AMCO)
	
7-hydroxy-4,8-dimethylcoumarin (HDMCO)	7-allyloxy-4,8-dimethylcoumarin (ADMCO)
	
5,7-dihydroxy-4-methylcoumarin (5,7-DHMCO)	5,7-diallyloxy-4-methylcoumarin (5,7-DAMCO)
	
7,8-dihydroxy-4-methylcoumarin (7,8-DHMCO)	7,8-diallyloxy-4-methylcoumarin (7,8-DAMCO)
	
6-Chloro-7-hydroxy-4-methylcoumarin (CHMCO)	6-Chloro-7-allyloxy-4-methylcoumarin (CAMCO)
	

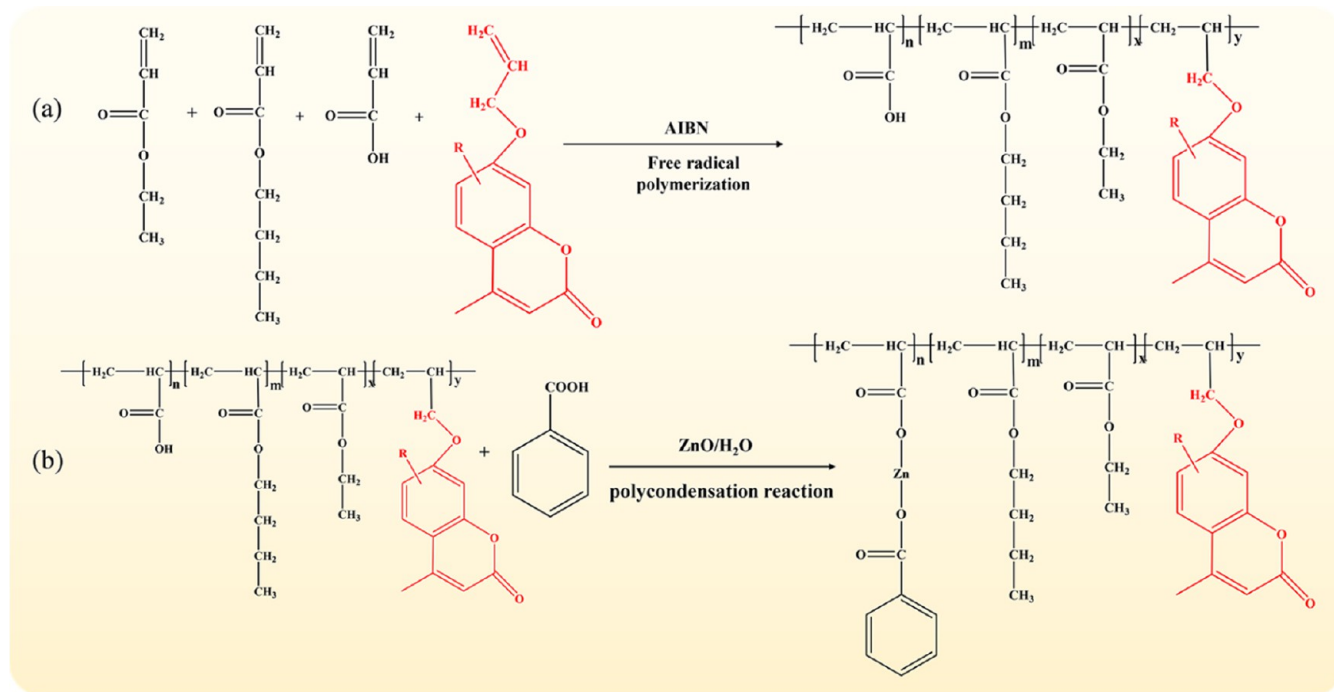
2.2. Synthesis of ZAR-Coumarin Coatings. **2.2.1. Synthesis of Coumarin Derivatives Containing Hydroxyl Groups.** Using different phenolic compounds and ethyl acetoacetate as raw materials, together with various catalysts (ZrCl_4 and H_2SO_4), a range of coumarin compounds with hydroxyl groups were produced using the Pechmann condensation process (Table 1).

Using ZrCl_4 as a catalyst and resorcinol, 2-methylresorcinol, and phloroglucinol as phenolic compounds, three coumarin compounds containing hydroxyl groups were prepared.³⁰ Briefly, 50 mmol of phenolic compound (resorcinol, 2-methylresorcinol, or phloroglucinol) was dissolved in 50 mmol of ethyl acetoacetate under ultrasound, and then 20 mmol of ZrCl_4 was added under stirring at 600 rpm. After reaction for 40 min, 100 g of crushed ice was added to the reaction. The solid product was collected by filtrations, then recrystallized using a mixed solvent of hot ethanol and water (V/V = 9:1) to obtain coumarin derivative containing hydroxyl groups. The obtained compounds were

7-hydroxy-4-methylcoumarin (HMCO), 7-hydroxy-4,8-dimethylcoumarin (HDMCO), and 5,7-dihydroxy-4-methylcoumarin (5,7-DHMCO), respectively.

Using H_2SO_4 as a catalyst and pyrogallol and 4-chlororesorcinol as phenolic compounds, two additional coumarin compounds containing hydroxyl groups were prepared.³¹ Briefly, 15 mL of concentrated H_2SO_4 was added to a three-necked flask, which was then cooled in an ice–water bath. Slowly, 50 mmol of 4-chlororesorcinol and 75 mmol of ethyl acetoacetate were added dropwise under stirring, with the resulting system allowed to react for 6 h while stirring at 1000 rpm. The resulting solution was then poured onto 100 g of crushed ice under vigorous stirring to precipitate the product. The solid product was collected by filtration, then redissolved in a 5 wt % NaOH solution, which was then extracted with ethyl acetate, and the aqueous layer collected. The aqueous layer was then acidified with 10 wt % H_2SO_4 to pH = 5 while cooling in an ice water bath, with the precipitate collected

Scheme 1. Synthesis Scheme of (a) Acrylic Prepolymers and (b) Zinc Acrylate Resins Containing Coumarin Units



by vacuum filtration and then washed with cold water. A recrystallization was performed in 75% ethanol, yielding coumarin compounds containing a hydroxyl structure denoted as 7,8-dihydroxy-4-methylcoumarin (7,8-DHMC) and 6-chloro-7-hydroxy-4-methylcoumarin (CHMCO), respectively. The chemical structures of coumarin compounds containing hydroxyl were confirmed by ¹H NMR (Figure S1) and infrared (IR) spectroscopy (Figure S2).

2.2.2. Synthesis of Coumarin Compounds with Allyl Groups. Coumarin derivatives containing allyl groups were synthesized using the Williamson reaction.³² Briefly, 32 mmol of a coumarin compound containing hydroxyl groups, 48 mmol of K₂CO₃, 48 mmol of 3-bromopropene, 0.9 mmol of KI, and 180 mL of acetone were added to a 250 mL three-necked flask equipped with a condenser tube and reacted at 60 °C under stirring (1000 rpm) for 24 h. After the reaction, the solid was collected by filtration and washed with acetone. The collected produce consisted of coumarin derivatives with allyl groups (Table 2). The chemical structures of the obtained coumarin derivatives were confirmed by proton nuclear magnetic resonance (¹H NMR) (Figure S3) and Fourier transform infrared spectroscopy (FTIR) spectroscopy (Figure S4).

2.2.3. Synthesis of ZAR-Coumarin Resins and Coatings. The coumarin derivatives with allyl groups were radically copolymerized with acrylic ester (ethyl acrylate and butyl acrylate) and acrylic acid to obtain acrylic prepolymers containing coumarin units (Scheme 1a). Taking AMCO as the functional monomer for example, prepolymer (40 wt %) was added to *n*-butanol and xylene (*n*-butanol/xylene mass ratio = 1:4) in a three-neck flask (250 mL). The flask was then fitted with a condenser, immersed in an 80 °C oil bath, the reaction heated under reflux while stirring at 1000 rpm. Next, stoichiometric amounts of acrylic monomers, including AMCO and AIBN (0.8% of the total monomer stoichiometry by weight), were introduced over 2 h under continuous stirring. After refluxing for 4 h at 1000 rpm, acrylic prepolymers containing coumarin units AMCO-*X* (*X* = 0.5, 1, 3, 5, 10, where *X* is the mole fraction of AMCO to total monomer expressed as a percentage) were obtained.

Next, ZAR-coumarin resins were obtained through a polycondensation reaction (Scheme 1b). Briefly, stoichiometric amounts of prepolymer, ZnO, and benzoic acid were added to a three-neck flask equipped with a stirrer and a condenser. Subsequently, xylene, *n*-butanol, and butyl acetate were added as solvents (60% of zinc acrylate

resin, wt %) and water (0.5% of total mass, wt %). The reaction was then carried out at 85 °C for 6 h under constant stirring (1000 rpm), resulting in the formation of a zinc acrylate resin containing AMCO.

The corresponding zinc acrylate resins are denoted as ZAR-AMCO-*X* (*X* = 0.5, 1, 3, 5, and 10, where *X* is the mole fraction of AMCO to total monomer expressed as a percentage).

ZAR-ADMCO, ZAR-CAMCO, ZAR-5,7-DAMCO, and ZAR-7,8-DAMCO and acrylate resins without coumarin units (ZAR, for comparison) were prepared by a similar method.

Table 3 shows the molar amounts of EA, BA, AA, the coumarin compound, ZnO, and benzoic acid used to prepare the various resins. The mole fractions of the coumarin compound are 0, 0.5, 1, 3, 5, and 10%, respectively.

2.3. Characterization. **2.3.1. Structural Characterization.** FTIR spectra were collected from 500 to 4000 cm⁻¹ on a Tensor II (Bruker) infrared spectrometer using the KBr disk method, with a blank KBr disk used to collect the background spectrum. ¹H NMR spectra were obtained on a Bruker BioSpin NMR spectrometer. Emission spectra of coatings were measured on a fluorescence spectrometer (Hitachi F-4600), with slit widths for emission (em) and excitation (ex) set at 5.0 and 5.0 nm, respectively. The voltage was 400 V, the scan speed was 12,000 nm/min, and the excitation wavelength was 350 nm. Absorbance spectra were collected by using a ultraviolet–visible (UV–visible) spectrophotometer (TU-1901, Persee). A NETZSCH STA 449 F3 Jupiter instrument (Germany) was used for simultaneous TG-DTG thermal analysis. Samples were heated from room temperature to 780 °C at 15 °C/min under a nitrogen flow (20 mL/min). Blank calibration was performed by using an empty Al₂O₃ crucible. A LEXT 3D measuring laser microscope OLS4000 (Olympus, Japan) was used to examine the surface morphology of the coatings. An artificial climate chamber with fluorescent lamps (RXZ-380C), Sanyo MLS-3750 autoclave (Japan), and constant temperature oscillation incubator (HZQ-F160) were employed for conducting antibacterial and antifungal tests.

2.3.2. Self-Polishing Performance. Each coating's capacity for self-polishing was evaluated through an experimental setup that involved dynamic paddling simulation, as illustrated in Figure 1. Briefly, 10 g of resin was applied onto a 350 × 100 × 3 mm³ ABS board, covering an area of 900 mm² (90 × 100 mm²), and allow to dry naturally. The thickness of the coatings was measured with a thickness gauge and

Table 3. Composition (Mole Ratio) of the Zinc Acrylate Resins Containing Coumarin Units^a

zinc acrylate resin	EA	BA	AA	coumarin compound (mole fractions)	ZnO	benzoic acid
ZAR	6	2	2	0	2	2
ZAR-AMCO-0.5	6	2	2	0.05 (0.5%)	2	2
ZAR-AMCO-1	6	2	2	0.101 (1%)	2	2
ZAR-AMCO-3	6	2	2	0.31 (3%)	2	2
ZAR-AMCO-5	6	2	2	0.53 (5%)	2	2
ZAR-AMCO-10	6	2	2	1.11 (10%)	2	2
ZAR-ADMCO-0.5	6	2	2	0.05 (0.5%)	2	2
ZAR-ADMCO-1	6	2	2	0.101 (1%)	2	2
ZAR-ADMCO-3	6	2	2	0.31 (3%)	2	2
ZAR-ADMCO-5	6	2	2	0.53 (5%)	2	2
ZAR-ADMCO-10	6	2	2	1.11 (10%)	2	2
ZAR-CAMCO-0.5	6	2	2	0.05 (0.5%)	2	2
ZAR-CAMCO-1	6	2	2	0.101 (1%)	2	2
ZAR-CAMCO-3	6	2	2	0.31 (3%)	2	2
ZAR-CAMCO-5	6	2	2	0.53 (5%)	2	2
ZAR-CAMCO-10	6	2	2	1.11 (10%)	2	2
ZAR-5,7-DAMCO-0.5	6	2	2	0.05 (0.5%)	2	2
ZAR-5,7-DAMCO-1	6	2	2	0.101 (1%)	2	2
ZAR-5,7-DAMCO-3	6	2	2	0.31 (3%)	2	2
ZAR-5,7-DAMCO-5	6	2	2	0.53 (5%)	2	2
ZAR-5,7-DAMCO-10	6	2	2	1.11 (10%)	2	2
ZAR-7,8-DAMCO-0.5	6	2	2	0.05 (0.5%)	2	2
ZAR-7,8-DAMCO-1	6	2	2	0.101 (1%)	2	2
ZAR-7,8-DAMCO-3	6	2	2	0.31 (3%)	2	2
ZAR-7,8-DAMCO-5	6	2	2	0.53 (5%)	2	2
ZAR-7,8-DAMCO-10	6	2	2	1.11 (10%)	2	2

^aNote: ZAR-CAMCO-10 could not be successfully synthesized under the same conditions.

adjusted to approximately 400 μm . The coated ABS board was then fixed on a dynamic simulation device filled with natural seawater, which rotated at 80 rpm/min. The experiment was conducted at 20 $^{\circ}\text{C}$ for 5 cycles, with 10 days representing one cycle. Samples of ABS board were removed every 10 days, cleaned with deionized water, then dried to a consistent weight in an oven set to 40 $^{\circ}\text{C}$. The resin coating surface's capacity for self-renewal was studied by periodically weighing the resin coating with an analytical balance and observing the changes in the

coatings' surface morphology by laser confocal focusing. The following formula was used to compute the resin coating's mass loss

$$\text{Mass loss(\%)} = \left(\frac{W_0 - W_t}{W_0} \right) \times 100\%$$

where W_0 denotes the initial mass of the coating and W_t denotes the weight of the coating after t days of the paddling experiment. For every sample, three parallel samples were created, and the average value was used to calculate the coating's mass loss.

2.3.3. Antibacterial Properties of ZAR-Coumarin Coatings. The effectiveness of the ZAR-coumarin coatings against bacteria were evaluated through the plate counting method, employing *Escherichia coli* (Gram-negative bacteria, DH5a) and *Bacillus* (Gram-positive bacteria, ZT-1, Ocean) selected as the test strains in this work. All materials used were sterilized in an autoclave and under UV light. Cleaned glass coverslips ($\Phi 16$ mm) were coated with ZAR-coumarin resins and allowed to dry naturally in air. Next, the coated glass coverslips were immersed in seawater for different periods of time (0, 10, 20, or 30 days). Next, they were removed from seawater and allowed to dry naturally at room temperature. Cultured bacteria suspensions were diluted to 10^{18} CFU/mL using a sterile NaCl solution (0.9 wt %). Next, the coated coverslips, which had been immersed in seawater for varying durations, were transferred into a 6-well plate. Each well was filled with 10^{18} CFU/mL of bacterial solution (5 mL), and the plate was incubated at 37 $^{\circ}\text{C}$ for 18 h. Subsequently, the cultured samples were removed from the bacterial solutions, with any excess liquid removed from the coating surface using a sterile cotton swab, after which the coating surfaces were washed several times with 5 mL of 0.9 wt % NaCl solution to obtain a rinsed bacterial solution. The bacterial solution was then diluted 100 times using a stepwise 10-fold dilution procedure. Next, 100 μL of the diluted bacterial liquid was applied uniformly on an agar plate surface, after which the plate was put in an incubator set at a constant temperature (37 $^{\circ}\text{C}$) for 15 h to observe the growth of the bacteria.

2.3.4. Antialgae Properties of ZAR-Coumarin. In this work, *Chlorella pyrenoidosa* (*C. pyrenoidosa*) and *Nitzschia closterium* (*N. closterium*) were selected to evaluate the inhibitory effect of the coatings on algae growth and algae settlement on the coatings. Algae were cultured in an artificial climate chamber (light/dark cycle of 12:12 h) using f/2 media. A previous study reported good absorption of light by algae at a wavelength of 680 nm.³⁴ The f/2 medium was used to dilute the initial optical density values of the algae, which were diluted to approximately $\text{OD}_{680} = 0.1$. Before the experiment, zinc acrylic resin was coated onto clean glass ($40 \times 40 \times 2$ mm³) and the coatings allowed to dry naturally. Subsequently, the coatings were immersed in seawater for 0, 10, 20, or 30 days. The

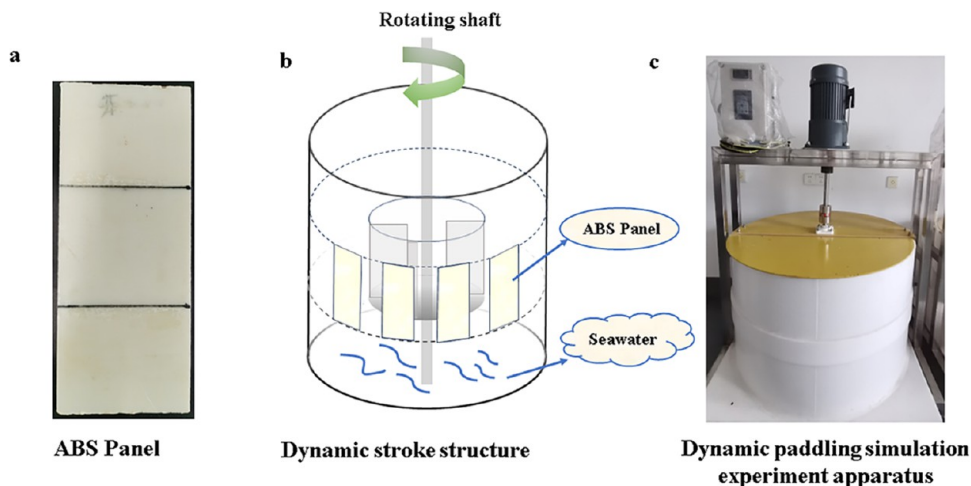


Figure 1. (a) ABS panel, (b) schematic diagram, and (c) experimental setup for the dynamic simulation experiment.³³

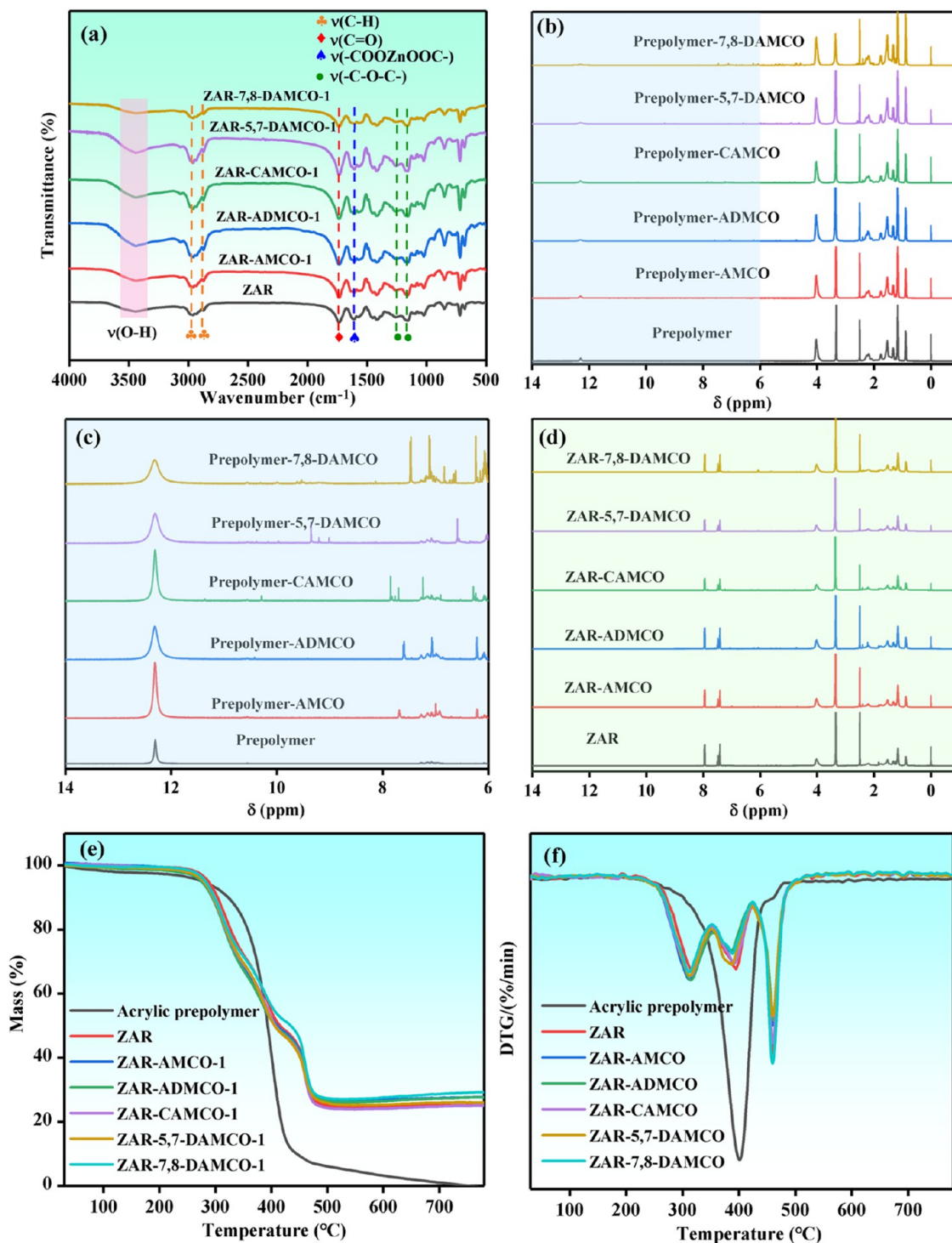


Figure 2. (a) FTIR spectra zinc acrylate resins containing coumarin units ((1 mol %)); (b, c) ^1H NMR spectra for the prepolymers (with (c) showing zoomed-in plots); (d) ^1H NMR spectra of zinc acrylate resins containing coumarin units; and (e, f) TGA and (f) DTG curves of the acrylic prepolymer and zinc acrylate resins containing coumarin units.

coatings were then removed from seawater and allowed to naturally dry in air at room temperature until further use.

The primary focus was to evaluate the impact of ZAR-coumarin coatings on algae growth. Specific experimental steps were as follows: Coatings that had been previously immersed in seawater for different times were placed in $70 \times 40 \text{ mm}^2$ glass containers containing 40 mL of a dilute algae suspension. Afterward, the containers were placed in an incubator and cultured for 7 days. The absorbance of the algal solution at 680 nm was measured every 24 h by using a UV–visible spectrophotometer. The effect of the coating on algae growth was

assessed by the change in the absorbance in the algae solution. To intuitively understand the effect of ZAR-coumarin coatings on algae growth, the relative inhibition rate was calculated based on the absorbance of the algae solution on the seventh day, using the following formula

$$\text{relative algae inhibition rate(\%)} = \frac{N_0 - N_s}{N_0} \times 100\%$$

where N_0 and N_s represent the number of algae on blank glass slides and coated glass slides after 7 days of algal growth in solution, respectively.

Next, the effect of the coating surface on algae settlement was evaluated by measuring the number of algae attached to the coatings under different conditions (dark and natural conditions). The specific experimental details are as follows: The coatings, which had not previously been immersed in seawater, were placed into glass containers measuring $70 \times 40 \text{ mm}^2$, then immersed in 40 mL of algae solution (approximately 10^5 cells/mL) and cultured for 7 days under both natural light and dark conditions, respectively. After culturing, samples were washed carefully with sterile seawater to remove any unattached algae. Subsequently, the coatings were placed in 10 mL of sterile seawater, subjected to 5 min ultrasound treatment, then scraped to remove any surface-adhered algae. Finally, the algae solution was collected, and its absorbance at 680 nm measured. In order to visualize the effect of resin coating on algae attachment, the relative inhibition rate was calculated as follows:

$$\text{relative alga einhibition rate(\%)} = \frac{N_1 - N_2}{N_1} \times 100\%$$

where N_1 represents the number of algae attached to blank glass slides and N_2 represents the number of algae attached to the surface of the ZAR-coumarin coatings.

2.3.5. Antifouling Performance in a Marine Environment. The antifouling performance of the coatings in a marine environment were conducted according to the national standard "Method for testing antifouling panels in shallow submergence" (GB/T 5370–2007). The specific experimental details are as follows: The PVC board was lightly sanded with 120 mesh sandpaper to enhance its roughness, followed by wiping with ethanol. Then 75 g of zinc acrylic resin was coated onto a PVC board measuring $300 \times 150 \times 6 \text{ mm}^3$. The test board was then submerged in seawater at Qingdao Port ($36^\circ 04' \text{N}$, $120^\circ 19' \text{E}$) and Rongcheng Sea ($37^\circ 23' \text{N}$, $122^\circ 42' \text{E}$) at a depth of about 1 m and routinely inspected. To examine biofouling adherence on the surface, the board was periodically removed from seawater to observe.

2.4. Evaluation of Molecular Planarity. Quantum chemical calculations were performed using Gaussian16 software. The configurations of coumarin derivatives were fully optimized by density functional theory (DFT) with wB97XD functional and def2TZVPP basis set,³⁵ including empirical dispersion correction for the large systems of monomers and the ZAR-coumarin. To estimate the impact the molecular coplanarity on the fluorescence data, the span of deviation from plane (SDP) was calculated using the Multiwfn 3.8 program³⁶ and the corresponding structures drawn by the VMD program.³⁷ The SDP delivers the deviation of different blocks in the structure from the plane, as described by eqs 1 and 2

$$d_i^s = \frac{A_{x_i} + B_{y_i} + C_{z_i} + D}{\sqrt{A^2 + B^2 + C^2}} \quad (1)$$

$$\text{SDP} = d_{\text{max}}^s - d_{\text{min}}^s \quad (2)$$

where d_i^s is the distance between atom i and the fitting plane, d_{max}^s and d_{min}^s denote the maximum and minimum values of d^s for the studied atoms, respectively.

A low value of SDP indicates a high planarity of the structure and a low deviation from the fitted plane. The SDP of coumarin units in the monomers and corresponding ZAR-coumarin coatings is useful for exploring the relationship between molecular structure and fluorescence properties.

3. RESULTS AND DISCUSSION

3.1. Structural Characterization and Thermodynamic Analysis. The chemical structures of the various ZAR-coumarin resins were analyzed by FTIR spectroscopy, as shown in Figures 2a and S5. In Figure 2a, the peak near 3440 cm^{-1} belongs to an O–H stretching vibration.³⁸ The absorption peaks located around 2980 and 2879 cm^{-1} were attributed to C–H stretching vibrations in the $-\text{CH}_2$ and $-\text{CH}_3$ groups, respectively.

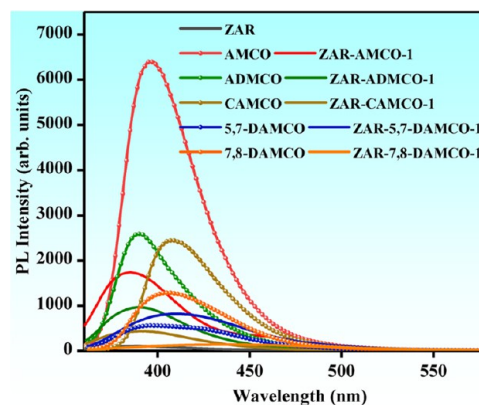


Figure 3. Fluorescence emission spectra of various coumarin compounds and ZAR-coumarin resins (1 mol % coumarin monomer).

Table 4. Fluorescence Emission Maxima (λ_{max}) of the Coumarin Compounds and the Corresponding ZAR-Coumarin Resins

coumarin compound	λ_{max} (nm)	resin	λ_{max} (nm)
AMCO	398	ZAR-AMCO-1	382
ADMCO	392	ZAR-ADMCO-1	388
CAMCO	410	ZAR-CAMCO-1	386
5,7-DAMCO	418	ZAR-5,7-DAMCO-1	406
7,8-DAMCO	405	ZAR-7,8-DAMCO-1	440

Furthermore, the peaks at about 1745 and 1615 cm^{-1} are assigned to stretching vibrations of $\text{C}=\text{O}$ and carboxylate $-\text{COOZnOOC}-$, respectively. The observation of these peaks demonstrate zinc acrylate resins were successfully prepared.³⁹ Moreover, the asymmetric and symmetric vibrations of $-\text{C}-\text{O}-\text{C}-$ gave characteristic absorption peaks at 1254 and 1163 cm^{-1} , respectively.⁴⁰ The FT-IR spectra for the zinc acrylate resins with different contents of coumarin compounds (Figure S5) were showed the same main peaks as resins containing a fixed amount of coumarin compound (Figure 2a), demonstrate the successful synthesis of zinc acrylate resins with both different types of coumarin units and different amounts of specific coumarin units.

Figure 2b–d displays the ^1H NMR spectra of the prepolymers and zinc acrylate resins containing coumaric units. In the acrylate prepolymer, the characteristic proton signal of the $-\text{COOH}$ group appears around $\delta = 12.3 \text{ ppm}$, whereas the H atoms on RCOOCH_2R units appear at $\delta = 3\text{--}4 \text{ ppm}$.³³ For acrylate prepolymers containing coumarin structures, the additional signal around $\delta = 6\text{--}8 \text{ ppm}$ is due to hydrogen atoms in the aromatic rings of the coumarin units, indicating the successful grafting of coumarin compounds onto acrylate resin prepolymer. Moreover, the signals appearing at $\delta = 1\text{--}2.5 \text{ ppm}$ are due to hydrogen atoms in CH_3 and CH_2 units. In the zinc acrylate resins (Figure 2d), the signal at $\delta = 12.3 \text{ ppm}$ disappeared, while signals located at around $\delta = 7\text{--}9 \text{ ppm}$ and $\delta = 7\text{--}8 \text{ ppm}$ appeared corresponding to H in coumarin and benzoic acid, respectively. These findings demonstrate that zinc acrylate resins containing coumarin units were successfully synthesized.

The thermal properties of acrylic prepolymers without coumarin units (acrylic prepolymer) and ZAR-coumarin resins were next investigated, with results shown in Figures 2e,f, S6 and S7. It is found that the ZAR-coumarin resins showed similar thermal properties to ZAR. For the ZAR-coumarin resins, three

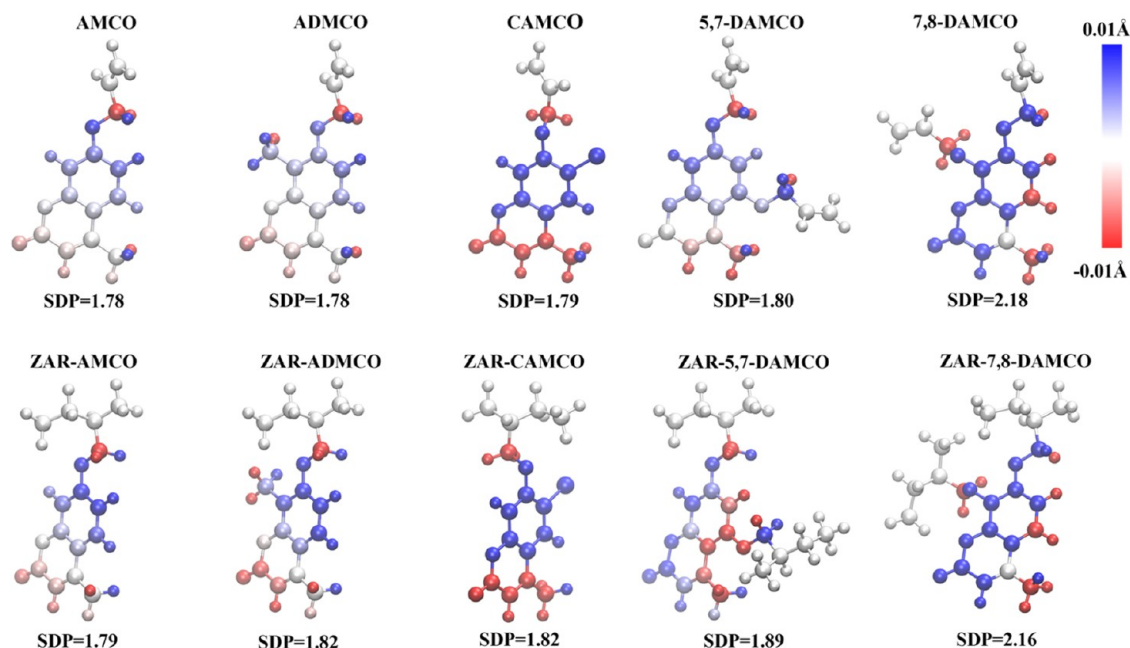


Figure 4. Span of deviation from plane (in Å) calculated for coumarin units in monomers and corresponding ZAR-coumarin at the ω B97XD/def2-TZVP level.

Table 5. Solid Content, Conversion Rate, and Acid Value of ZAR-Coumarin Resins^a

resins	ZAR	ZAR-AMCO-1	ZAR-ADMCO-1	ZAR-CAMCO-1	ZAR-5,7-DAMCO-1	ZAR-7,8-DAMCO-1
solid content (%)	38.715 $\pm$ 0.105	37.585 $\pm$ 0.375	35.935 $\pm$ 0.185	37.81 $\pm$ 0.17	33.45 $\pm$ 0.41	34.82 $\pm$ 0.2
conversion rate (%)	96.525 $\pm$ 0.263	93.963 $\pm$ 0.938	89.838 $\pm$ 0.463	94.525 $\pm$ 0.425	83.625 $\pm$ 1.025	87.05 $\pm$ 0.5
PAV (mg KOH/g)	102.07 $\pm$ 3.02	150.10 $\pm$ 5.41 (*)	161.11 $\pm$ 1.03 (*)	135.34 $\pm$ 1.69 (*)	186.48 $\pm$ 2.03 (*)	175.73 $\pm$ 4.35 (*)

^aNote: Reported values are the average of measurements on three independent samples; “ $\pm$ ” stands for error ($n = 3$). Brackets indicates statistical significance relative to ZAR (*, $P < 0.05$).

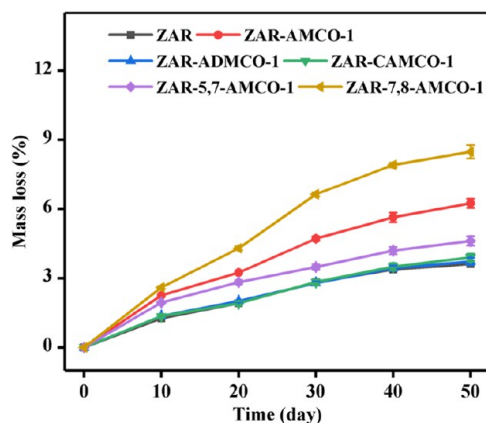


Figure 5. Mass loss of ZAR and ZAR-coumarin (1 mol %) coatings during self-polishing tests. Reported values are the average of three independent samples; error bars indicate the standard error ($n = 3$).

largest mass losses occurred at 315, 390, and 460 °C, whereas the poly(acrylic acid) prepolymer only showed two mass loss at 400 and 460 °C. Compared with acrylic acid prepolymers, the extra mass loss for the ZAR-coumarin resins observed at 315 °C is likely attributable to the decomposition of the carboxylate groups/uncomplexed groups in coumarin.⁴¹ The mass loss at about 390 °C may be attributed to the breakage of the C–C bonds in the main chain of the acrylic resin,⁴² while the mass loss at about 460 °C corresponds to breakage of the zinc acrylate-saturated polymer chains.⁴² Furthermore, as shown in Figure S7,

the glass transition temperature of ZAR-AMCO-1 is 46.09 °C, which is much higher than that of the zinc acrylate resin (10 °C),⁴³ further indicating that the zinc acrylate resin in this work has good stability.

3.2. Fluorescent Properties. The five coumarin monomers and five ZAR-coumarin resins show excellent fluorescent properties, as shown in Figures 3 and S8 and Table 4. The maximum emission peaks for AMCO, ADMCO, CAMCO, 5,7-DAMCO, and 7,8-DAMCO were located at 398, 392, 410, 418, and 405 nm, respectively. Meanwhile, the maximum emission peaks for ZAR-AMCO-1, ZAR-ADMCO-1, ZAR-CAMCO-1, and ZAR-5,7-DAMCO-1 blue-shifted to 382, 388, 386, and 406 nm, respectively. For, ZAR-7,8-DAMCO-1, a red-shift to 440 nm was observed. Similar trends were seen for all ZAR-coumarin resins prepared with different coumarin unit contents (Figure S8). The shifts seen in the maximum emission peaks with resin formation likely results from changes in the molecular coplanarity of the coumarin units.⁴⁴ A schematic representation of the molecular planarity is illustrated in Figure 4, where blue and red indicate the deviations above and below the plane, respectively. As can be seen, after grafting, the SDP value of the coumarin unit in ZAR-7,8-DAMCO-1 decreased compared with the corresponding monomer, indicating the coumarin units in ZAR-7,8-DAMCO-1 exhibit better planarity. This result is consistent with the observed red-shift phenomenon. In contrast, the SDP values of the coumarin units in the other four ZAR-coumarin structures increased compared with their monomers, showing a decrease in planarity. This decrease in planarity can be

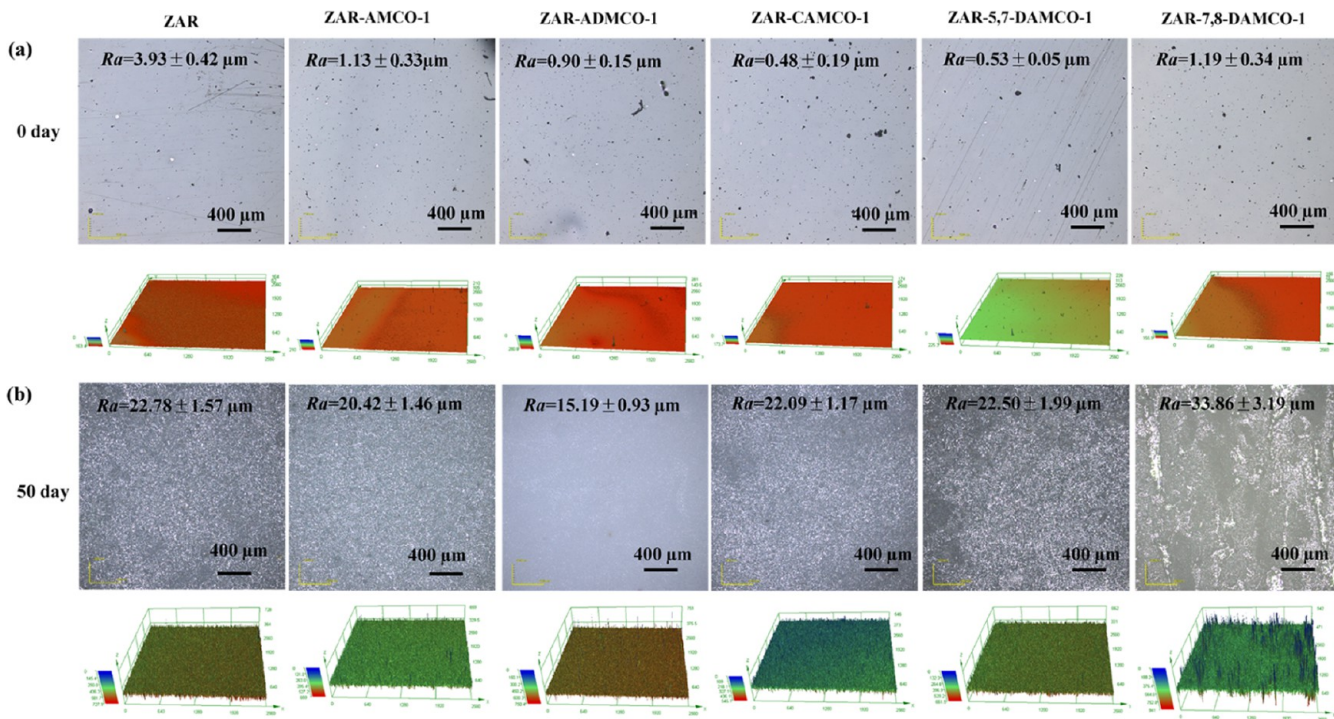


Figure 6. Two-dimensional (2D) and three-dimensional (3D) microscopy images of ZAR and ZAR-coumarin coatings (1 mol %). Surface (a) before and (b) after 5 cycles of dynamic simulation experiment. Note: Ra stands for the surface roughness of the corresponding resins.

Table 6. Inhibition of *E. coli* by ZAR-Coumarin (1 mol %) Coatings after Being Submerged for Different Times in Seawater^a

	Blank	ZAR	ZAR-AMCO-1	ZAR-ADMCO-1	ZAR-CAMCO-1	ZAR-5,7-DAMCO-1	ZAR-7,8-DAMCO-1
0 days							
10 days							
20 days							
30 days							

^aNote: Blank was a cleaned glass coverslip.

used to explain the blue shift in the emission wavelength of these four ZAR-coumarin resins.

3.3. Solid Content and Acid Value. Tables 5 and S1 summarize the test results for the solid content and acid value (AV) of the various ZAR-coumarin resins. According to the results, the solid content of the zinc acrylate resins is below the theoretical value of 40%. From Table 5, it can be seen that the

solid content of zinc acrylate resins containing double-substituted functional monomers (5,7-DAMCO and 7,8-DAMCO) was lower than that of zinc acrylate resins containing monosubstituted functional monomers (AMCO, ADMCO, CAMCO), which may be due to the spatial site resistance effect. For monosubstituted functional monomers, the effect of content changes on the solid content was not significant (Table S1). It is

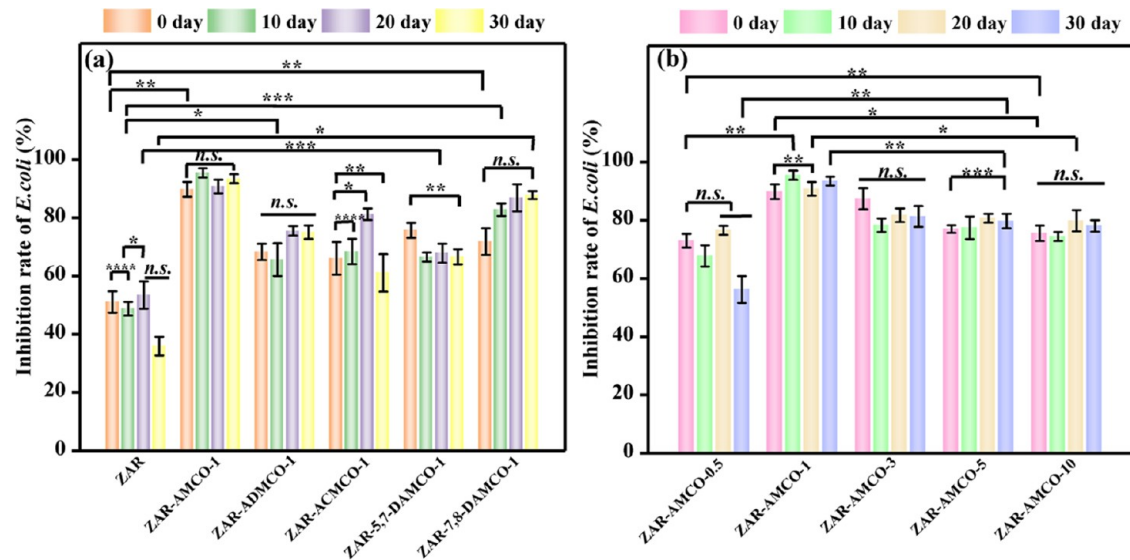


Figure 7. Inhibition rates for *E. coli* by (a) ZAR-coumarin (1 mol %) coatings and (b) ZAR-AMCO coatings with different AMCO contents after being submerged for different times in seawater. Reported values are the average of three independent samples; error bars indicate the standard error ($n = 3$). Significance levels are denoted as * ($P < 0.05$), ** ($P < 0.01$), *** ($P < 0.001$), **** ($P < 0.0001$), and n.s. (not significant).

Table 7. Inhibition of *E. coli* by ZAR-AMCO Coatings with Different AMCO Contents after Being Submerged for Different Times in Seawater^a

	Blank	ZAR-AMCO-0.5	ZAR-AMCO-1	ZAR-AMCO-3	ZAR-AMCO-5	ZAR-AMCO-10
0 day						
10 days						
20 days						
30 days						

^aNote: Blank was a cleaned glass coverslip.

worth noting that there is a negative correlation between the polymer acid value and the solid content, which can be attributed to benzoic acid and zinc oxide reacting to generate zinc benzoate that is not grafted to the acrylic copolymer. This results in more carboxyl groups on the acrylic copolymer, which results in its higher acid value and a lower solid content.³⁸ Simultaneously, the zinc acrylate resin's acid value can be improved to 150–190 mg KOH/g via the addition of the

coumarin-containing structures, thus ensuring a high rate of hydrolysis.

3.4. Self-Polishing Performance. The self-polishing ability of the ZAR-coumarin resins were assessed via dynamic simulation experiments. The mass loss of the resin coating with cycle testing is shown in Figure 5. All resins showed the highest mass loss after the first cycle, which could be attributed to the attack of seawater on residual unreacted acid and low-molecular-

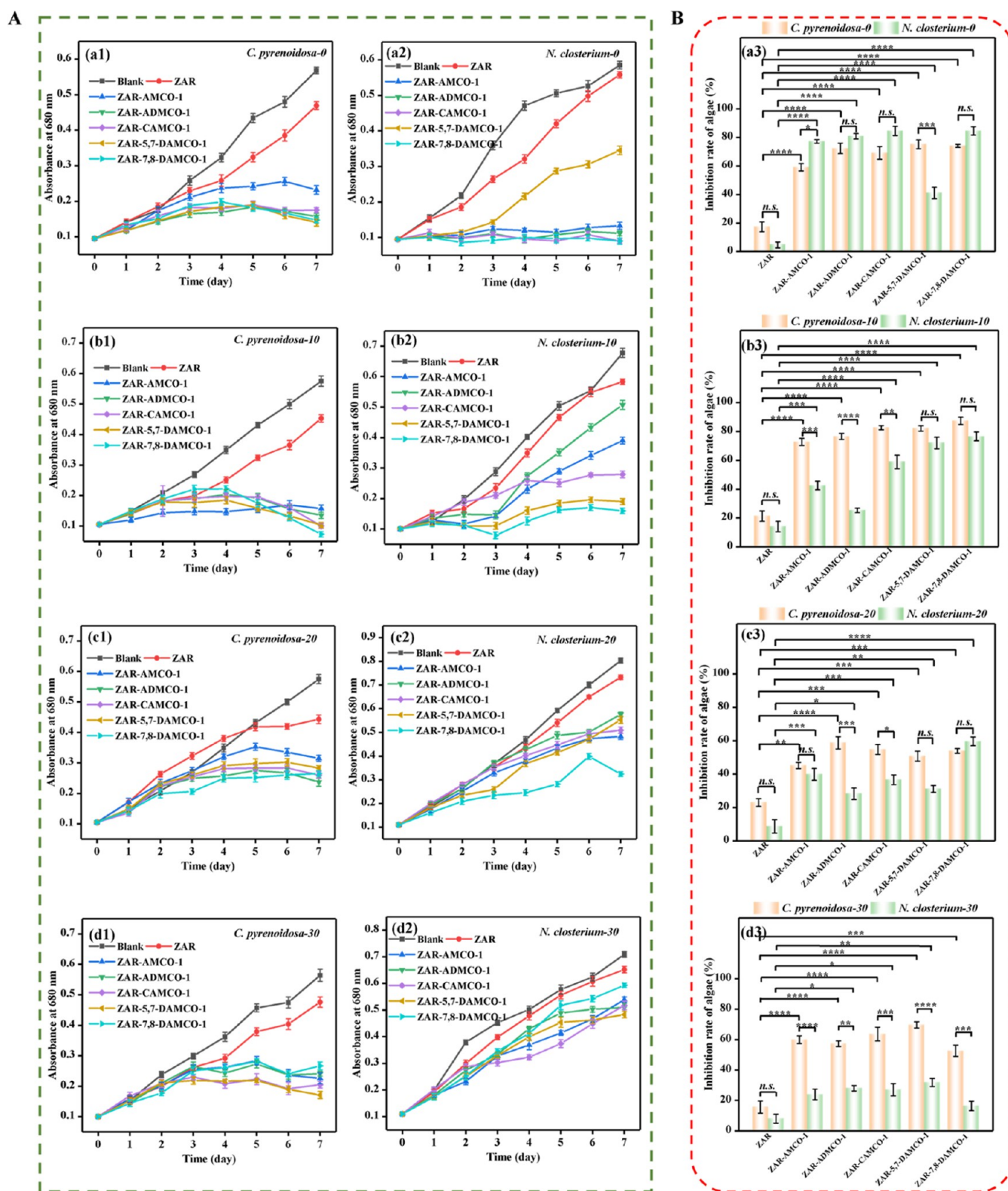


Figure 8. (A) Effect of ZAR-coumarin resins (1 mol % coumarin) on algae growth inhibition following immersion in seawater for different times and (B) corresponding algae inhibition rate after 7 days of growth. Reported values are the average of measurements on three independent samples; error bars indicate the standard error ($n = 3$). Significance levels are denoted as * ($P < 0.05$), ** ($P < 0.01$), *** ($P < 0.001$), **** ($P < 0.0001$), and n.s. (not significant). Note: Blank stands for the cleaned glass coverslip.

weight prepolymers.⁴⁵ The mass loss of the coatings gradually leveled off after the second and third cycles, which can be attributed to slow hydrolysis of the ester groups in the zinc acrylic resins.⁴⁶ The coating's mass loss stabilized after the fourth

cycle, indicating a very slow rate of resin hydrolysis.⁴⁷ In addition, the hydrolysis rate of ZAR-7,8-DAMCO-1 was higher than that of ZAR, which may be due to its larger steric hindrance due to the introduced coumarin units and incomplete hydrolysis

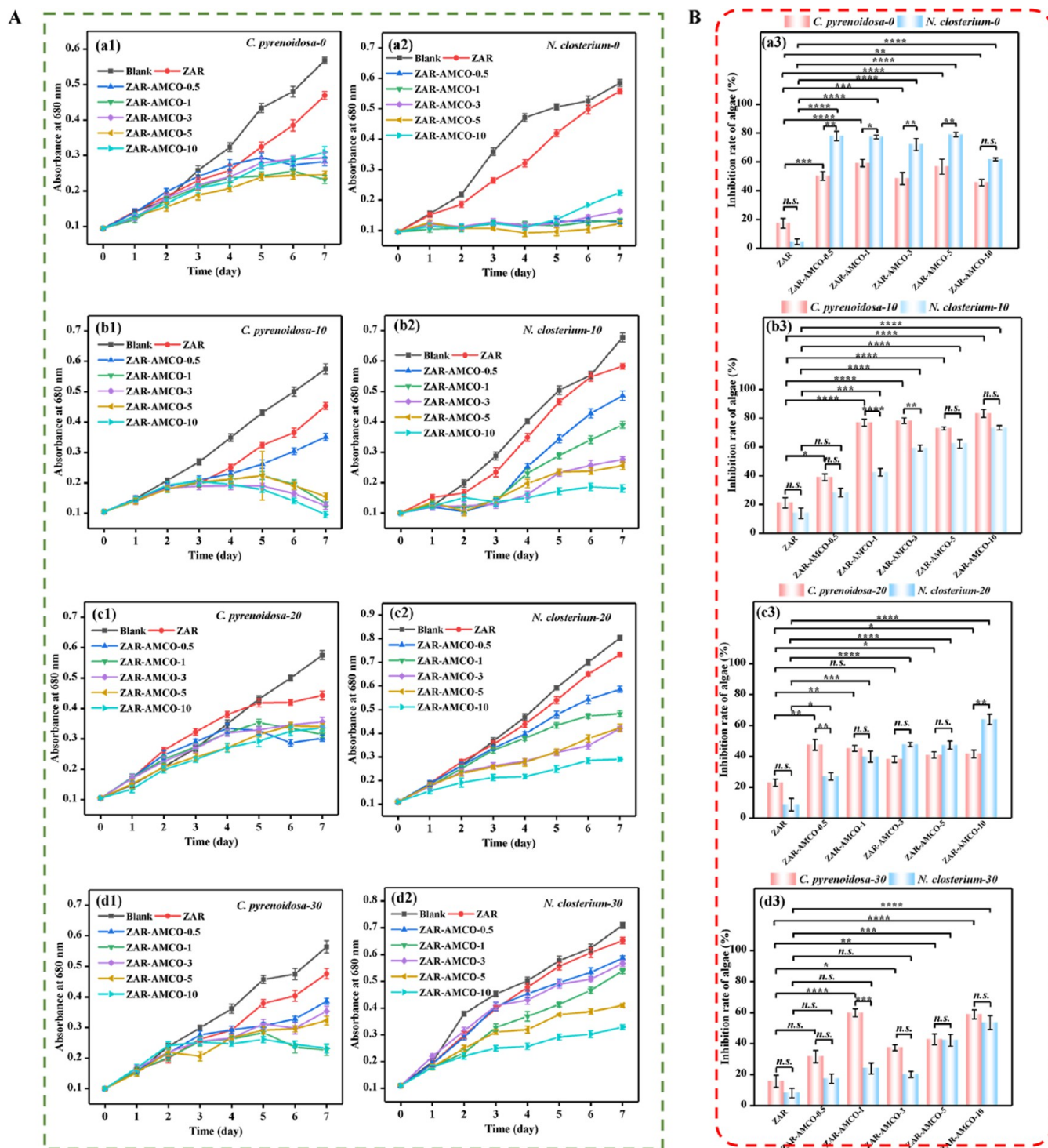


Figure 9. (A) Effect of ZAR-AMCO resins with different AMCO contents on algae growth inhibition after being submerged for different times in seawater and (B) corresponding algae inhibition rate after 7 days of growth. Reported values are the average of measurements on three independent samples; error bars indicate the standard error ($n = 3$). Significance levels are denoted as * ($P < 0.05$), ** ($P < 0.01$), *** ($P < 0.001$), **** ($P < 0.0001$), and n.s. (not significant). Note: Blank stands for the cleaned glass coverslip.

processes. At the same time, ZAR-ADMCO-1 and ZAR-CAMCO-1 showed a much lower mass loss, indicating that the coatings may have extended antifouling performance.

Subsequently, the ZAR-coumarin coating surface morphologies were examined by using laser confocal microscopy before and after the dynamic simulation experiment (Figure 6 and Table S2). Before the experiment, the surfaces were relatively

flat ($R_a < 2$), with no wrinkles or cracks being observed. This indicates that the ZAR-coumarin resins had good film-forming properties (Figure 6a). After five cycles of dynamic simulation experiments, the ZAR-coumarin coating's surface roughness increased to about $20 \mu\text{m}$, with some protrusions seen on the surface, which was likely due to the rapid and inhomogeneous hydrolysis of the ester groups on the coating surface.⁴⁸ Except

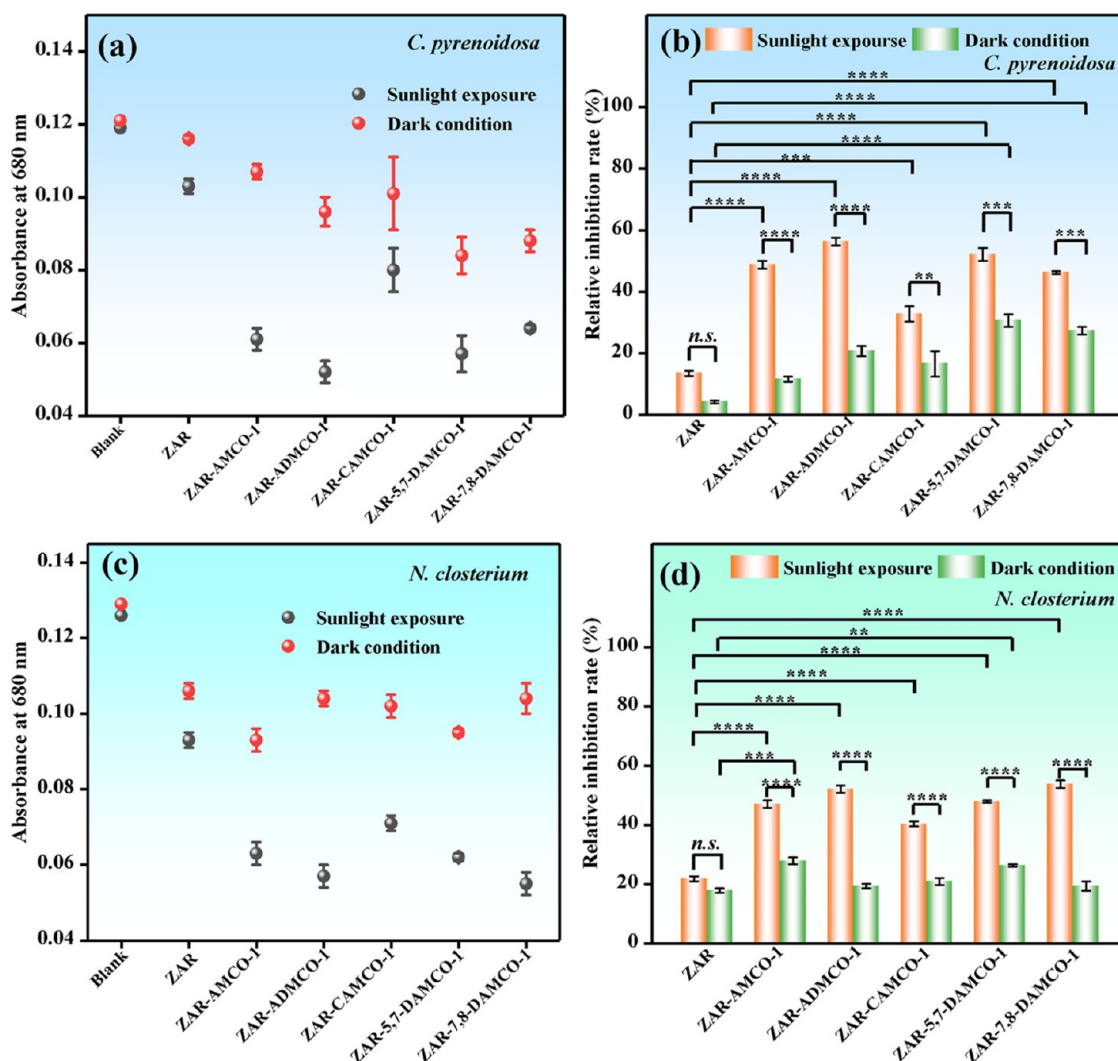


Figure 10. Settlement results for (a, b) *C. pyrenoidosa* and (c, d) *N. closterium* on ZAR-coumarin coatings (1 mol %) under dark conditions and sunlight exposure. Reported values are the average of measurements on three independent samples; error bars indicate the standard error ($n = 3$). Significance levels are denoted as * ($P < 0.05$), ** ($P < 0.01$), *** ($P < 0.001$), **** ($P < 0.0001$), and n.s. (not significant). Note: Blank indicates a cleaned glass coverslip.

for ZAR-7,8-DAMCO-1, all other ZAR-coumarin coatings remained flat and crack-free, indicating that the overall hydrolysis of the ZAR-coumarin coatings was uniform and stable (Figure 6b). The exception was the ZAR-7,8-DAMCO-1 coating, which showed severe wrinkles on the surface and increased roughness, which may be due to the large spatial potential resistance and incomplete reaction of 7,8-DAMCO. Table S2 demonstrates that the ZAR-coumarin surfaces generally remained flat and did not change significantly after the test cycles, with the coumarin content having little bearing on the self-polishing performance.

3.5. Antibacterial Properties. Using the plate spreading method with *E. coli* (Gram-negative bacteria) as the target strain, the inhibition of adhesion of the bacteria on the coated surfaces after being in seawater for different periods was investigated. After being exposed to seawater for 0, 10, 20, or 30 days, the ZAR-coumarin resins exhibited resistance to bacterial adhesion, as shown in Table 6 and Figure 7a. Figure 7a illustrates that ZAR-coumarin coatings have a higher resistance to bacterial adhesion (>60%) than ZAR (<50%). This difference can be attributed to the nature of coumarins. Coumarins are natural

antimicrobial agents, that can impact bacterial growth and survival by rupture of cell membranes,⁴⁹ damage to DNA,²⁹ inhibition of enzyme activities,⁵⁰ and other mechanisms. Among the coatings, the ZAR-AMCO-1 coating showed the highest inhibition efficiency (more than 90%) against *E. coli*. Further, the effect of the AMCO content on the inhibitory efficiency against *E. coli* was investigated, as shown in Table 7 and Figure 7b. The results show that all ZAR-AMCO can effectively inhibit the attachment and growth of *E. coli* (inhibition rate >65%). In addition, the inhibition effect remained almost unchanged with increasing exposure time in seawater (except for ZAR-AMCO-1). The greatest inhibitory effect of 90% was realized when the AMCO molar content was 1 mol %. We also investigated the initial coating resistance to *Bacillus* (Ocean) adhesion, with results shown Figures S9 and S10. The data showed that the ZAR-coumarin coatings can effectively inhibit the adhesion of *Bacillus* with an inhibition efficiency of more than 90%. This indicates that the ZAR-coumarin coatings have a good broad-spectrum antibacterial activity.

3.6. Algae Inhibition Properties. The algae inhibition properties of the ZAR-coumarin coatings were tested by using *C.*

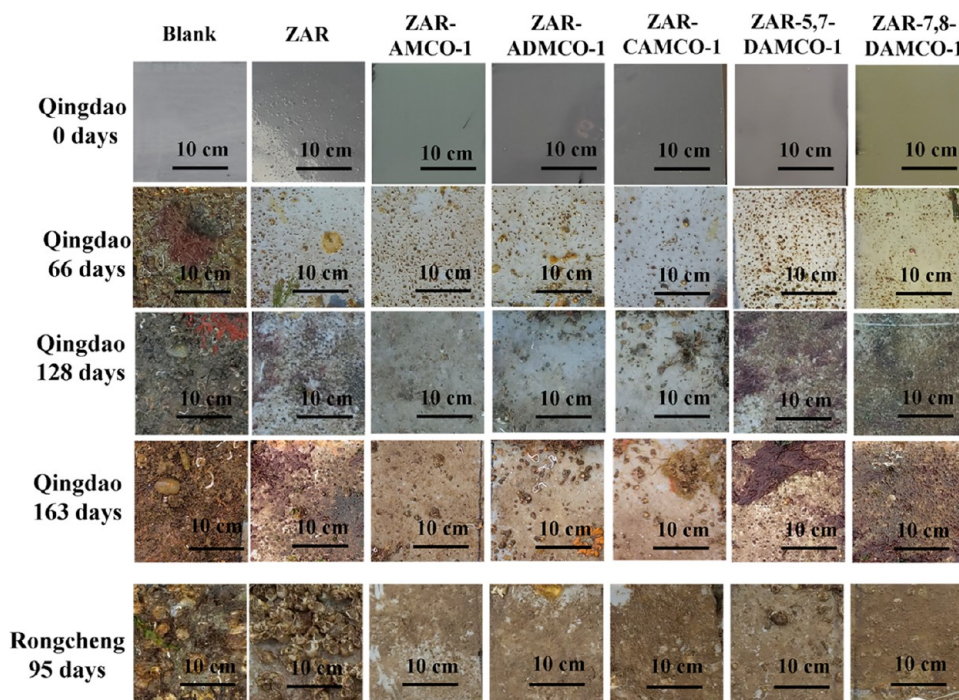


Figure 11. Optical images of ZAR-coumarin coatings containing different coumarin derivatives (1 mol %) after immersion in seawater for different times. Note: Blank indicates an uncoated PVC panel.

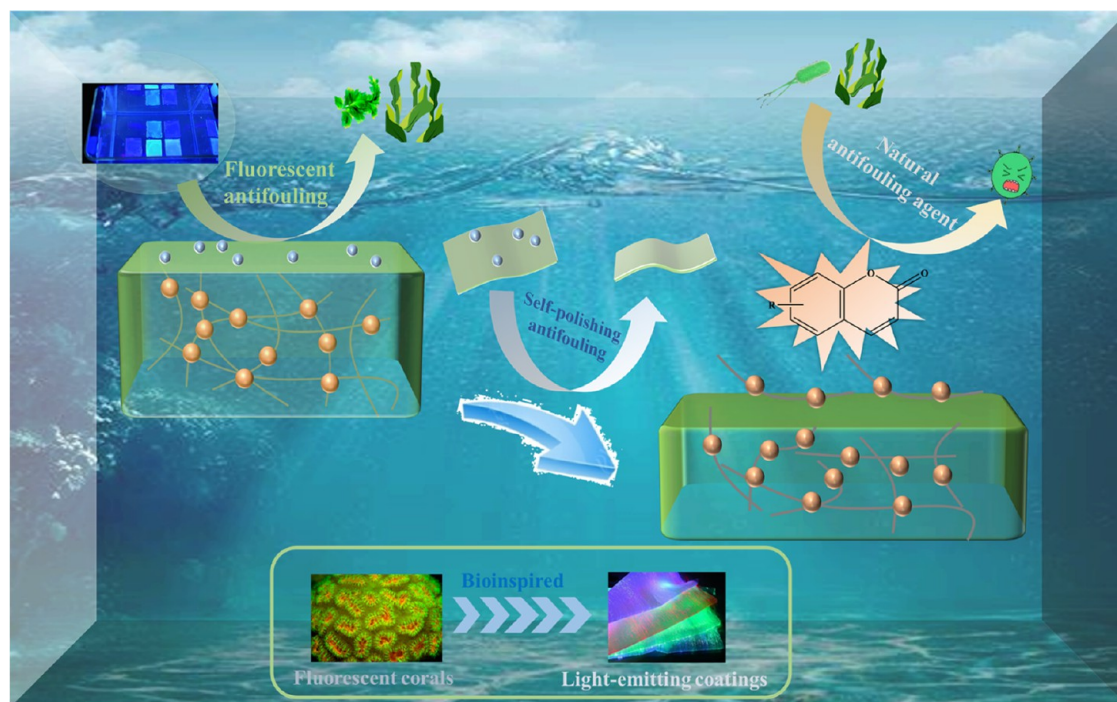


Figure 12. Antifouling mechanisms of ZAR-coumarin coatings.

pyrenoidosa (green algae) and *N. closterium* (diatom) as the test algae species, as shown in Figure 8. The ZAR-coumarin coatings also show good inhibitory efficiency (~80%) against *C. pyrenoidosa* and *N. closterium* after immersion in seawater for 0 and 10 days. After 20 and 30 days, the inhibiting efficiency decreased to 50 and 30%, respectively. Overall, the ZAR-coumarin coatings were more effective in preventing the growth of algae compared with a control ZAR coating, which can be attributed to the fact that coumarin can act as an inhibitor of

DNA topoisomerase (Topo II, EC 5.99.1.3) and inhibits cell growth.^{28,51,52}

Taking ZAR-AMCO as a representative example, we investigated the effect of coumarin content in the coatings on algae growth was investigated. From Figure 9, the coumarin content in ZAR-AMCO was shown to exert little effect on the inhibition of *C. pyrenoidosa* and *N. closterium*. All ZAR-AMCO coatings exhibited a good inhibitory impact on algae, with an inhibition rate of approximately 70% for *C. pyrenoidosa* and *N.*

closterium after immersion in algae-containing seawater for up to 10 days. Nevertheless, after immersion for 20 and 30 days, the inhibitory efficiency for *C. pyrenoidosa* and *N. closterium* dropped to around 40 and 30%, respectively.

3.7. Algae Sedimentation Properties. Algae settle on coating surfaces, forming a biofilm that plays a crucial role in the development of biofouling. Accordingly, we investigated the impact of ZAR-coumarin coatings on preventing algae settlement, as shown in Figure 10. The results indicate that ZAR-coumarin coatings are better than ZAR coatings in preventing algae settlement under the same experimental conditions. This may be ascribed to the fact coumarin is a natural antibacterial agent that can effectively block ATPase activity in cells, which in turn inhibits cell growth.^{28,51} In addition, under sunlight conditions, the inhibition rate for algae attachment on ZAR-coumarin coatings was significantly higher than that under dark conditions (<30%). This can be attributed to the fluorescent antifouling properties of ZAR-coumarin coatings, where algae will preferentially photosynthesize rather than settle with light exposure.^{21–23} In summary, fluorescent antifouling, the bioactivity of coumarin, and the self-renewal properties of the ZAR-coumarin coatings all contribute to inhibiting algae settlement.

3.8. Marine Antifouling Performance. We conducted an offshore hanging board test in the sea area of Qingdao Harbor (36°04'N, 120°19'E, June–November 2023) and Rongcheng Sea (37°23'N, 122°42'E, July–October 2023), both located in Shandong Province, to assess the antifouling properties of the ZAR-coumarin coatings in real marine environments, as shown in Figure 11. In Qingdao Harbor, during the first 90 days (June–July 2023), no fouling was observed on the ZAR surface. During the rapid growth period of marine organisms (July–August 2023), the presence of some microorganisms and silt was evident on the ZAR coating's surface. After 90 and 128 days of submersion in the ocean, silt and small microorganisms adhered to the ZAR coating, while no obvious fouling organisms were on the ZAR-coumarin coatings (except ZAR-S,7-DAMCO-1). After 163 days, the surface of the ZAR-AMCO-1, ZAR-ADMCO-1, and ZAR-CAMCO-1 coatings had begun to be colonized by tiny organisms such as barnacles, though there was still not too much fouling pressure on the ZAR-coumarin coatings compared to the ZAR coating, indicating that ZAR-AMCO-1, ZAR-ADMCO-1, and ZAR-CAMCO-1 offered stable antifouling properties under static conditions. Similarly, in the Rongcheng sea area, after 95 days of immersion in seawater, the surfaces of blank and ZAR were fully covered by barnacle attachment, with an attachment area reaching 100%. In contrast, the surface of ZAR-coumarin was only covered by silt without any obvious microbial attachment, indicating that ZAR-coumarin still exhibited excellent antifouling performance. Furthermore, as observed in Figure S11, the antifouling efficacy of ZAR-coumarin coatings decreased when the coumarin amount surpassed 5 mol %, which may be related to the self-absorption and quenching of the fluorescence intensity (Figure S8). Overall, the findings of the marine field hanging board show that ZAR-coumarin coatings perform well in practical maritime environments.

3.9. Antifouling Mechanism. Building upon the above findings, we can propose the potential antifouling mechanism(s) used by ZAR-coumarin coatings, as shown in Figure 12. First, fluorescent antifouling exerts a certain inhibitory effect on algae settlement, primarily due to light's influence on the spatial competition between chloroplasts and holdfast-like structures

within the membrane. Algae will prioritize photosynthesis instead of sedimentation under high light conditions.^{21,23,25} Second, coumarin and its derivatives can affect cell survival and proliferation by acting as an inhibitor of DNA topoisomerase (Topo II), which can obstruct DNA replication and transcription and cause a buildup of damage to DNA, cell cycle arrest, or apoptosis.⁵² Moreover, they can stop ATPase activity, and thus stop DNA supercoiling and cell development.²⁸ Hence, as natural antibacterial agents, coumarin and its derivatives are crucial in preventing the proliferation of marine microorganisms on ZAR-coumarin coatings. Finally, zinc acrylate resin is a typical self-polishing antifouling resin in which the antifouling ability mainly arises from the hydrolysis of the ester group in the side chain. Simultaneously, when an external shear stress acts, the hydrolyzed coating surface will progressively erode and peel off in a layer-by-layer manner, in the process removing organisms adhered to the coating surface and enabling constant renewal of the coating surface.^{8,9} Moreover, the Zn²⁺ released during hydrolysis penetrates the cells, disrupting mitochondrial function and inducing cell apoptosis.⁵⁴ This research provides a new strategy toward high-performance marine antifouling coatings.

4. CONCLUSIONS

Inspired by the natural antifouling strategy of fluorescent corals, a series of fluorescent ZAR-coumarin coatings was prepared that offer impressive antibacterial and antialgae properties, making them suitable for application in natural marine environments. The excellent antifouling capabilities of the ZAR-coumarin coatings were due to fluorescent antifouling and natural antibacterial properties of the coumarin units, along with the self-polishing antifouling properties typical of zinc acrylate resins. Results demonstrate that zinc acrylate resins containing coumarin units offer a multimechanism synergistic approach for suppressing marine biofouling.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.biomac.4c01574>.

FTIR and ¹H NMR spectra for coumarin compounds containing hydroxyl and coumarin compounds containing allyl groups were confirmed in this work; fluorescence emission spectra of ZAR were confirmed in this work; solid content and acid value of ZAR-coumarin coatings; surface morphology of ZAR-coumarin coatings; inhibition effect for *Bacillus* of ZAR; optical images of ZAR-coumarin coatings after immersing in seawater for different times, and additional information in the work (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Zhiming Zhang – Frontiers Science Center for Deep Ocean Multispheres and Earth System, and Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China; Sanya Oceanographic Institution, Ocean University of China; Sanya Oceanographic Laboratory, Sanya 572024, China;

✉ orcid.org/0000-0001-9825-8859; Email: zzmcyj@ouc.edu.cn

Liangmin Yu – Frontiers Science Center for Deep Ocean Multispheres and Earth System, and Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China; Sanya Oceanographic Institution, Ocean University of China; Sanya Oceanographic Laboratory, Sanya 572024, China; Email: yuyan@ouc.edu.cn

Authors

Jianwei Hou – Frontiers Science Center for Deep Ocean Multispheres and Earth System, and Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China

Guobo Chen – Frontiers Science Center for Deep Ocean Multispheres and Earth System, and Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China; orcid.org/0000-0002-5601-5442

Xinghai Hao – Frontiers Science Center for Deep Ocean Multispheres and Earth System, and Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China

Jiali Xu – Frontiers Science Center for Deep Ocean Multispheres and Earth System, and Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China

Geoffrey I. N. Waterhouse – School of Chemical Sciences, The University of Auckland, Auckland 1142, New Zealand; orcid.org/0000-0002-3296-3093

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.biomac.4c01574>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (No. 42376206), the MacDiarmid Institute for Advanced Materials, Nanotechnology, and the Dodd Walls Centre for Photonic and Quantum Technologies.

REFERENCES

- (1) Hou, J.; Xu, J.; Hao, X.; Waterhouse, G. I. N.; Zhang, Z.; Yu, L. Photoelectric antifouling performance of molybdenum disulfide/graphite carbon nitride/polyaniline coatings under visible light. *Appl. Surf. Sci.* **2024**, 657, No. 159794.
- (2) Li, S.; Feng, K.; Li, J.; Li, Y.; Li, Z.; Yu, L.; Xu, X. Marine antifouling strategies: Emerging opportunities for seawater resource utilization. *Chem. Eng. J.* **2024**, 486, No. 149859.
- (3) Cahill, P. L.; Moodie, L. W. K.; Hertzner, C.; Pinori, E.; Pavia, H.; Hellio, C.; Brimble, M. A.; Svenson, J. Creating New Antifoulants Using the Tools and Tactics of Medicinal Chemistry. *Acc. Chem. Res.* **2024**, 57, 399–412.
- (4) Hong, H.; Lv, J.; Deng, A.; Tang, Y.; Liu, Z. A review of experimental Assessment Processes of material resistance to marine and freshwater biofouling. *J. Environ. Manage.* **2024**, 357, No. 120766.
- (5) Xu, G.; Liu, X.; Liu, P.; Pranantyo, D.; Neoh, K.; Kang, E. Arginine-Based Polymer Brush Coatings with Hydrolysis-Triggered Switchable Functionalities from Antimicrobial (Cationic) to Antifouling (Zwitterionic). *Langmuir* **2017**, 33, 6925–6936.
- (6) Chen, J.; Bai, W.; Jian, R.; Lin, Y.; Zheng, X.; Wei, F.; Lin, Q.; Lin, F.; Xu, Y. Molecular structure design of polybenzoxazines with low surface energy and low modulus for marine antifouling application. *Prog. Org. Coat.* **2024**, 187, No. 108165.

- (7) Chen, J.; Jian, R.; Yang, K.; Bai, W.; Huang, C.; Lin, Y.; Zheng, B.; Wei, F.; Lin, Q.; Xu, Y. Urushiol-based benzoxazine copper polymer with low surface energy, strong substrate adhesion and antibacterial for marine antifouling application. *J. Cleaner Prod.* **2021**, 318, No. 128527.
- (8) Hu, K.; Li, M.; You, C.; Zhang, Y.; Xu, Z.; Xu, Y.; Fan, L.; Yi, Y.; Chu, Y. Novel indole-based self-polishing environmentally friendly acrylic antifouling coatings. *Prog. Org. Coat.* **2024**, 190, No. 108384.
- (9) Sha, J.; Liu, X.; Chen, R.; Yu, J.; Liu, Q.; Liu, J.; Zhu, J.; Liu, P.; Li, R.; Wang, J. Surface hydrolysis-anchored eugenol self-polishing marine antifouling coating. *J. Colloid Interface Sci.* **2023**, 637, 67–75.
- (10) Qiu, H.; Feng, K.; Gapeeva, A.; Meurisch, K.; Kaps, S.; Li, X.; Yu, L.; Mishra, Y. K.; Adelung, R.; Baum, M. Functional polymer materials for modern marine biofouling control. *Prog. Polym. Sci.* **2022**, 127, No. 101516.
- (11) Xu, G.; Liu, P.; Pranantyo, D.; Neoh, K.; Kang, E. Dextran- and Chitosan-Based Antifouling, Antimicrobial Adhesion, and Self-Polishing Multilayer Coatings from pH-Responsive Linkages-Enabled Layer-by-Layer Assembly. *ACS Sustainable Chem. Eng.* **2018**, 6, 3916–3926.
- (12) Ma, C.; Zhang, W.; Zhang, G.; Qian, P. Environmentally Friendly Antifouling Coatings Based on Biodegradable Polymer and Natural Antifoulant. *ACS Sustainable Chem. Eng.* **2017**, 5, 6304–6309.
- (13) Tian, S.; Jiang, D.; Pu, J.; Sun, X.; Li, Z.; Wu, B.; Zheng, W.; Liu, W.; Liu, Z. A new hybrid silicone-based antifouling coating with nanocomposite hydrogel for durable antifouling properties. *Chem. Eng. J.* **2019**, 370, 1–9.
- (14) Zhang, J.; Wu, M.; Peng, P.; Liu, J.; Lu, J.; Qian, S.; Feng, J. Self-Defensive Antifouling Zwitterionic Hydrogel Coatings on Polymeric Substrates. *ACS Appl. Mater. Interfaces* **2022**, 14, 56097–56109.
- (15) Chen, L.; Duan, Y.; Cui, M.; Huang, R.; Su, R.; Qi, W.; He, Z. Biomimetic surface coatings for marine antifouling: Natural antifoulants, synthetic polymers and surface microtopography. *Sci. Total Environ.* **2021**, 766, No. 144469.
- (16) Jin, H.; Tian, L.; Bing, W.; Zhao, J.; Ren, L. Bioinspired marine antifouling coatings: Status, prospects, and future. *Prog. Mater. Sci.* **2022**, 124, No. 100889.
- (17) Zhao, Z.; Pan, M.; Qiao, C.; Xiang, L.; Liu, X.; Yang, W.; Chen, X. Z.; Zeng, H. Bionic Engineered Protein Coating Boosting Anti-Biofouling in Complex Biological Fluids. *Adv. Mater.* **2023**, 35, No. 2208824.
- (18) Selim, M. S.; El-Safty, S. A.; et al. Progress in biomimetic leverages for marine antifouling using nanocomposite coatings. *J. Mater. Chem. B* **2020**, 8, 3701–3732.
- (19) Sridhar, A.; Sekar, R. K.; Manikandan, D. B.; Arumugam, M.; Veeran, S.; Ramasamy, T. Activity profile of innate immune-related enzymes and bactericidal of freshwater fish epidermal mucus extract at different pH. *Environ. Sci. Pollut. Res.* **2021**, 28, 33914–33926.
- (20) Zhao, H.; Sun, Q.; Deng, X.; Cui, J. Earthworm-Inspired Rough Polymer Coatings with Self-Replenishing Lubrication for Adaptive Friction-Reduction and Antifouling Surfaces. *Adv. Mater.* **2018**, 30, No. 1802141.
- (21) Cao, S.; Wang, J.; Chen, D. Influence of Illumination on Settlement of Diatom *Navicula* sp. *Microb. Ecol.* **2011**, 62, 931–940.
- (22) Guo, H.; Song, L.; Hu, J.; Lin, T.; Li, X.; Yu, H.; Cheng, D.; Hou, Y.; Zhan, X.; Zhang, Q. Enhanced antifouling strategy with a strong synergistic effect of fluorescent antifouling and contact bacteriostasis using 7-amino-4-methylcoumarin. *Chem. Eng. J.* **2021**, 420, No. 127676.
- (23) Li, Z.; Dong, Y.; Mi, B.; Tang, Y.; Häussler, M.; Tong, H.; Dong, Y.; Lam, J. W. Y.; Ren, Y.; Sung, H. H. Y.; et al. Structural Control of the Photoluminescence of Silole Regioisomers and Their Utility as Sensitive Regiodiscriminating Chemosensors and Efficient Electroluminescent Materials. *J. Phys. Chem. B* **2005**, 109, 10061–10066.
- (24) Mao, Q.; Liu, S.; Xiong, Y.; Hu, D.; Huang, L.; Fang, Z.; Jiang, H.; Wang, H.; Li, J.; Mao, S.; Wang, G. Advanced Marine Antifouling Hydrogels Based on 7-Amino-4-methylcoumarin Fluorescence Driven by Rare-Earth Phosphorescence. *ACS Appl. Mater. Interfaces* **2023**, 15, 57582–57592.

- (25) Jin, H.; Bing, W.; Jin, E.; Tian, L.; Jiang, Y. Bioinspired PDMS–Phosphor–Silicone Rubber Sandwich-Structure Coatings for Combating Biofouling. *Adv. Mater. Interfaces* **2020**, *7*, No. 1901577.
- (26) Sun, X.-Y.; Liu, T.; et al. Synthesis and application of coumarin fluorescence probes. *RSC Adv.* **2020**, *10*, 10826–10847.
- (27) Peng, X. M.; Damu, G. L.; Zhou, C. Current developments of coumarin compounds in medicinal chemistry. *Curr. Pharm. Des.* **2013**, *19*, 3884–3930.
- (28) Feng, D.; Zhang, A.; Yang, Y.; Yang, P. Coumarin-containing hybrids and their antibacterial activities. *Arch. Pharm.* **2020**, *353*, No. c1900380.
- (29) Menezes, J. C.; Diederich, M. Translational role of natural coumarins and their derivatives as anticancer agents. *Future Med. Chem.* **2019**, *11*, 1057–1082.
- (30) Sharma, G. V. M.; Reddy, J. J.; Lakshmi, P. S.; Krishna, P. R. An efficient ZrCl₄ catalyzed one-pot solvent free protocol for the synthesis of 4-substituted coumarins. *Tetrahedron Lett.* **2005**, *46*, 6119–6121.
- (31) Tan, Y.-L.; Zou, X.-W.; Z, W.-H. Synthesis of 8 (or 6) -chloro-7-hydroxy-4-methylcoumarin and their derivatives. *Chin. J. Synth. Chem.* **2009**, *17*, 312–315.
- (32) Zhang, X. Masters; Nanjing University of Science and Technology, 2009.
- (33) Hou, J.; Hao, X.; Xu, J.; Waterhouse, G. I. N.; Ni, C.; Zhang, Z.; Yu, L. Fluorescent zinc acrylate resins containing coumarin structures with enhanced antifouling performance. *Prog. Org. Coat.* **2024**, *194*, No. 108597.
- (34) Hou, J.; Zhao, H.; Zhang, Z.; Yu, L.; Yan, X. The antifouling tris-(8-hydroxyquinoline) aluminum: Titanium dioxide coatings under visible light. *Surf. Coat. Technol.* **2023**, *468*, No. 129743.
- (35) Lin, Y.-S.; Li, G.; Mao, S.; Chai, A. J. Long-Range Corrected Hybrid Density Functionals with Improved Dispersion Corrections. *J. Chem. Theory Comput.* **2013**, *9*, 263–272.
- (36) Lu, T.; Chen, A. F. Multiwfn: A Multifunctional Wavefunction Analyzer. *J. Comput. Chem.* **2012**, *33*, 580–592.
- (37) Humphrey, W.; Dalke, A.; Schulten, K. VMD: visual molecular dynamics. *J. Mol. Graphics* **1996**, *14*, 33–38.
- (38) Yangpeng, L.; Xiang, L. E.; Jinxing, S.; Bajin, C.; Shengpeng, W.; Quan, L.; Yongyuan, R.; Xiaoli, Z.; Qinghua, Z. Study on the Preparation and Performance of Acrylic Zinc Resin. *Chem. React. Eng. Technol.* **2022**, *38*, 515–524.
- (39) Deng, Z.; Wang, Y.; Zhang, D.; Chen, C. 3D printing technology meets marine biofouling: A study on antifouling resin for protecting marine sensors. *Addit. Manuf.* **2023**, *73*, No. 103697.
- (40) Feng, K.; Ni, C.; Yu, L.; Zhou, W.; Li, X. Synthesis and evaluation of acrylate resins suspending indole derivative structure in the side chain for marine antifouling. *Colloids Surf., B* **2019**, *184*, No. 110518.
- (41) Sebastian, N.; George, B.; Mathew, B. Metal complexes of poly(acrylic acid): synthesis, characterization and thermogravimetric studies. *Polym. Degrad. Stab.* **1998**, *60*, 371–375.
- (42) Chen, R.; Li, Y.; Yan, M.; Sun, X.; Han, H.; Li, J.; Wang, J.; Liu, L.; Takahashi, K. Synthesis of hybrid zinc/silyl acrylate copolymers and their surface properties in the microfouling stage. *RSC Adv.* **2016**, *6*, 13858–13866.
- (43) Chen, R. Ph.D.; Harbin Engineering University, 2017.
- (44) Zhang Yuan, W.; Zhang, Y. Nonconventional macromolecular luminogens with aggregation-induced emission characteristics. *J. Polym. Sci., Part A: Polym. Chem.* **2017**, *55*, 560–574.
- (45) Chen, S.; Ma, C.; Zhang, G. Biodegradable polymer as controlled release system of organic antifoulant to prevent marine biofouling. *Prog. Org. Coat.* **2017**, *104*, 58–63.
- (46) Ni, C.; Feng, K.; Li, X.; Zhao, H.; Yu, L. Study on the preparation and properties of new environmentally friendly antifouling acrylic metal salt resins containing indole derivative group. *Prog. Org. Coat.* **2020**, *148*, No. 105824.
- (47) Zhou, W.; Wang, Y.; Ni, C.; Yu, L. Preparation and evaluation of natural rosin-based zinc resins for marine antifouling. *Prog. Org. Coat.* **2021**, *157*, No. 106270.
- (48) Sha, J.; Yu, J.; Chen, R.; Liu, Q.; Liu, J.; Zhu, J.; Liu, P.; Li, R.; Wang, J. Eco-friendly self-polishing antifouling coating via eugenol ester hydrolysis. *Prog. Org. Coat.* **2022**, *172*, No. 107077.
- (49) Yang, L.; Ding, W.; Xu, Y.; Wu, D.; Li, S.; Chen, J.; Guo, B. New Insights into the Antibacterial Activity of Hydroxycoumarins against *Ralstonia solanacearum*. *Molecules* **2016**, *21*, No. 468.
- (50) Yang, E. B.; Zhao, Y. N.; Zhang, K.; Mack, P. Daphnetin, One of Coumarin Derivatives, Is a Protein Kinase Inhibitor. *Biochem. Biophys. Res. Commun.* **1999**, *260*, 682–685.
- (51) Liu, H.; Ren, Z.; Wang, W.; Gong, J.; Chu, M.; Ma, Q.; Wang, J.; Lv, X. Novel coumarin-pyrazole carboxamide derivatives as potential topoisomerase II inhibitors: Design, synthesis and antibacterial activity. *Eur. J. Med. Chem.* **2018**, *157*, 81–87.
- (52) Yorgey, P.; Lee, J.; Kordel, J.; Vivas, E.; Warner, P.; Jebaratnam, D.; Kolter, R. Posttranslational Modifications in Microcin B17 Define an Additional Class of DNA Gyrase Inhibitor. *Proc. Natl. Acad. Sci. U.S.A.* **1994**, *91*, 4519–4523.
- (53) Sha, J.; Chen, R.; Yu, J.; Liu, Q.; Liu, J.; Zhu, J.; Liu, P.; Li, R.; Wang, J. Dynamic multi-level microstructured antifouling surfaces by combining quaternary ammonium modified GO with self-polishing copolymers. *Carbon* **2023**, *201*, 1038–1047.
- (54) Formigari, A.; Irato, P.; Santon, A. Zinc, antioxidant systems and metallothionein in metal mediated-apoptosis: Biochemical and cytochemical aspects. *Comp. Biochem. Physiol., Part C: Toxicol. Pharmacol.* **2007**, *146*, 443–459.



CAS BIOFINDER DISCOVERY PLATFORM™

CAS BIOFINDER HELPS YOU FIND YOUR NEXT BREAKTHROUGH FASTER

Navigate pathways, targets, and
diseases with precision

Explore CAS BioFinder



A Division of the
American Chemical Society